

CSCI2750 Homework 3

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1 Problem 1

1.1 Part (ai)

Dataset:

$$a = (1, 2), \quad b = (2, 1), \quad c = (0, 1), \quad d = (1, 0).$$

Mean:

$$\frac{a + b + c + d}{4} = \left(\frac{1 + 2 + 0 + 1}{4}, \frac{2 + 1 + 1 + 0}{4} \right) = (1, 1).$$

De-meaned dataset:

$$X = \begin{pmatrix} 1 - 1 & 2 - 1 \\ 2 - 1 & 1 - 1 \\ 0 - 1 & 1 - 1 \\ 1 - 1 & 0 - 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}.$$

1.2 Part (aii)

Labels: $y_a = y_b = +1$ and $y_c = y_d = -1$. Treat each sample and w as row vectors in $\mathbb{R}^{1 \times 2}$. Use the perceptron update rule

$$w \leftarrow w + y_i x_i \quad \text{if } y_i(x_i w^\top) \leq 0,$$

with initial point $w_0 = (0, 0)$ and the order (a, b, c, d) .

$$a' = (0, 1), \quad b' = (1, 0), \quad c' = (-1, 0), \quad d' = (0, -1).$$

1. Start with $w = (0, 0)$.
2. For a' : $y_a(a' w^\top) = +1 \cdot 0 = 0 \leq 0$, add a' to w .

$$w = (0, 0) + (0, 1) = (0, 1).$$

3. For b' : $y_b(b' w^\top) = +1 \cdot 0 = 0 \leq 0$, add b' to w .

$$w = (0, 1) + (1, 0) = (1, 1).$$

4. For c' : $y_c(c' w^\top) = -1 \cdot (-1) = 1 > 0$, no update.

5. For d' : $y_d(d' w^\top) = -1 \cdot (-1) = 1 > 0$, no update.

After one pass, the perceptron finds $w = (1, 1)$, which defines a separating hyperplane.

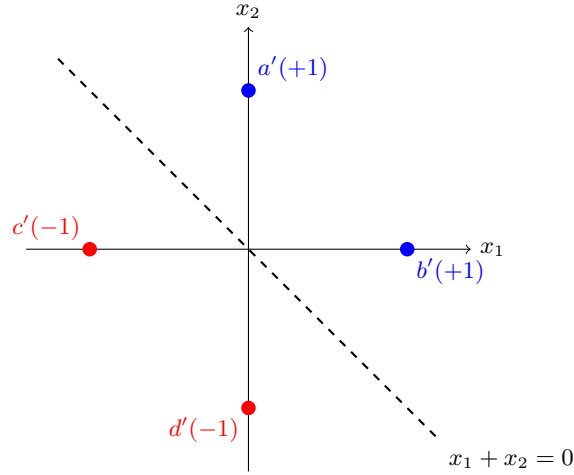
1.3 Part (aiii)

The de-meaned points are

$$a' = (0, 1), \quad b' = (1, 0), \quad c' = (-1, 0), \quad d' = (0, -1).$$

A separating hyperplane found in Part (aii) is

$$w = (1, 1), \quad xw^\top = 0 \iff x_1 + x_2 = 0.$$



For margin, we have

$$\gamma = \min_i \frac{y_i(x_i w^\top)}{\|w\|_2}.$$

Calculating $y_i(x_i w^\top)$ gives 1 for all four points, and $\|w\|_2 = \sqrt{2}$, hence

$$\gamma = \frac{1}{\sqrt{2}}.$$

This is the largest achievable margin for this symmetric dataset.

Conclusion From Part (aii), the basic perceptron gives $w = (1, 1)$, so the separating hyperplane is

$$xw^\top = 0 \iff x_1 + x_2 = 0.$$

Using this w , the margin is

$$\gamma = \min_i \frac{y_i(x_i w^\top)}{\|w\|_2} = \frac{1}{\sqrt{2}}.$$

From the geometric symmetry of the de-meaned dataset, the largest achievable margin is also $\frac{1}{\sqrt{2}}$. Therefore, the separator obtained in Part (aii) is a max-margin separator for this dataset.

1.4 Part (b)

From Figure 1, we have the following dataset (row-vector form):

Class 1: $x_1 = (-3, 1)$, $x_2 = (-2, 2)$,

Class 2: $x_3 = (0, 3)$, $x_4 = (3, 4)$,

Class 3: $x_5 = (-1, -4)$, $x_6 = (1, -2)$.

Let $w_1, w_2, w_3 \in \mathbb{R}^{1 \times 2}$ with initialization

$$w_1^0 = w_2^0 = w_3^0 = (0, 0).$$

Prediction rule:

$$\hat{y} = \arg \max_{k \in \{1, 2, 3\}} x w_k^\top.$$

(Tie-break rule: choose the smallest class index.) If $\hat{y} \neq y$, update:

$$w_y \leftarrow w_y + x, \quad w_{\hat{y}} \leftarrow w_{\hat{y}} - x,$$

and keep the remaining class weight unchanged.

Per-epoch comparison table (all samples):

Epoch	Sample	(w_1, w_2, w_3) state	Score vec (s_1, s_2, s_3)	True y	Pred $\hat{y}$	Mistake?
1	$x_1 = (-3, 1)$	$((0, 0), (0, 0), (0, 0))$	$(0, 0, 0)$	1	1	No
1	$x_2 = (-2, 2)$	$((0, 0), (0, 0), (0, 0))$	$(0, 0, 0)$	1	1	No
1	$x_3 = (0, 3)$	$((0, 0), (0, 0), (0, 0))$	$(0, 0, 0)$	2	1	Yes
1	$x_4 = (3, 4)$	$((0, -3), (0, 3), (0, 0))$	$(-12, 12, 0)$	2	2	No
1	$x_5 = (-1, -4)$	$((0, -3), (0, 3), (0, 0))$	$(12, -12, 0)$	3	1	Yes
1	$x_6 = (1, -2)$	$((1, 1), (0, 3), (-1, -4))$	$(-1, -6, 7)$	3	3	No
2	$x_1 = (-3, 1)$	$((1, 1), (0, 3), (-1, -4))$	$(-2, 6, -1)$	1	2	Yes
2	$x_2 = (-2, 2)$	$((-2, 2), (3, 2), (-1, -4))$	$(8, -2, -6)$	1	1	No
2	$x_3 = (0, 3)$	$((-2, 2), (3, 2), (-1, -4))$	$(6, 6, -12)$	2	1	Yes
2	$x_4 = (3, 4)$	$((-2, -1), (3, 5), (-1, -4))$	$(-10, 30, -19)$	2	2	No
2	$x_5 = (-1, -4)$	$((-2, -1), (3, 5), (-1, -4))$	$(6, -23, 17)$	3	3	No
2	$x_6 = (1, -2)$	$((-2, -1), (3, 5), (-1, -4))$	$(-4, -2, 7)$	3	3	No
3	$x_1 = (-3, 1)$	$((-2, -1), (3, 5), (-1, -4))$	$(5, -7, 5)$	1	1	No
3	$x_2 = (-2, 2)$	$((-2, -1), (3, 5), (-1, -4))$	$(4, 11, -6)$	1	2	Yes
3	$x_3 = (0, 3)$	$((-4, 1), (5, 3), (-1, -4))$	$(-3, 15, -12)$	2	2	No
3	$x_4 = (3, 4)$	$((-4, 1), (5, 3), (-1, -4))$	$(-2, 27, -19)$	2	2	No
3	$x_5 = (-1, -4)$	$((-4, 1), (5, 3), (-1, -4))$	$(0, -17, 17)$	3	3	No
3	$x_6 = (1, -2)$	$((-4, 1), (5, 3), (-1, -4))$	$(-6, -1, 7)$	3	3	No
4	$x_1 = (-3, 1)$	$((-4, 1), (5, 3), (-1, -4))$	$(13, -12, -1)$	1	1	No
4	$x_2 = (-2, 2)$	$((-4, 1), (5, 3), (-1, -4))$	$(10, -4, -6)$	1	1	No
4	$x_3 = (0, 3)$	$((-4, 1), (5, 3), (-1, -4))$	$(3, 9, -12)$	2	2	No
4	$x_4 = (3, 4)$	$((-4, 1), (5, 3), (-1, -4))$	$(-8, 27, -19)$	2	2	No
4	$x_5 = (-1, -4)$	$((-4, 1), (5, 3), (-1, -4))$	$(0, -17, 17)$	3	3	No
4	$x_6 = (1, -2)$	$((-4, 1), (5, 3), (-1, -4))$	$(-6, -1, 7)$	3	3	No

Hence, convergence is reached at Epoch 4 (the first zero-mistake epoch).

2 Problem 2

2.1 Part (a)

Find the equations of the hyperplanes h_1 and h_2 .

By observation:

- The line h_1 passes through points such as $(3, 8)$ and $(5, 2)$, so slope is

$$m_1 = \frac{2 - 8}{5 - 3} = -3.$$

Hence

$$\frac{x_2 - 8}{x_1 - 3} = -3 \iff x_2 = -3x_1 + 17 \iff h_1(x_1, x_2) = 3x_1 + x_2 - 17 = 0.$$

- The line h_2 passes through points such as $(2, 1)$ and $(4, 4)$, so slope is

$$m_2 = \frac{4 - 1}{4 - 2} = \frac{3}{2}.$$

Hence

$$\frac{x_2 - 1}{x_1 - 2} = \frac{3}{2} \iff x_2 = \frac{3}{2}x_1 - 2 \iff h_2(x_1, x_2) = 3x_1 - 2x_2 - 4 = 0.$$

2.2 Part (b)

Support vectors are points with smallest perpendicular distance to the hyperplane, equivalently smallest $|h(x)|$.

We use the formula for distance from a point to a hyperplane:

$$\frac{|ax_1 + bx_2 + c|}{\sqrt{a^2 + b^2}}.$$

For $h_1(x) = 3x_1 + x_2 - 17$:

- Blue points: $(2, 6) \mapsto |-5| = 5$, $(3, 4) \mapsto |-4| = 4$, $(2, 2.5) \mapsto |-8.5| = 8.5$. Shortest is $(3, 4)$.
- Red points: $(6, 2) \mapsto |3| = 3$, $(7, 6) \mapsto |10| = 10$, $(8, 4) \mapsto |11| = 11$. Shortest is $(6, 2)$.

So support vectors for h_1 are

$$(3, 4) \text{ (blue) and } (6, 2) \text{ (red)}.$$

Consider $h_2(x) = 3x_1 - 2x_2 - 4$:

- Blue points: $(2, 6) \mapsto |-10| = 10$, $(3, 4) \mapsto |-3| = 3$, $(2, 2.5) \mapsto |-3| = 3$. Shortest is $(3, 4)$ and $(2, 2.5)$.
- Red points: $(6, 2) \mapsto |10| = 10$, $(7, 6) \mapsto |5| = 5$, $(8, 4) \mapsto |12| = 12$. Shortest is $(7, 6)$.

So support vectors for h_2 are

$$(3, 4), (2, 2.5) \text{ (blue) and } (7, 6) \text{ (red)}.$$

2.3 Part (c)

For hyperplane $ax_1 + bx_2 + c = 0$, normal vector is (a, b) .

For h_1 : normal vector $(3, 1)$, so $\|w_1\| = \sqrt{10}$ and minimum $|h_1(x)| = 3$ (calculated in Part (b)).

$$\gamma_1 = \frac{3}{\sqrt{10}} \approx 0.949.$$

For h_2 : normal vector $(3, -2)$, so $\|w_2\| = \sqrt{13}$ and minimum $|h_2(x)| = 3$.

$$\gamma_2 = \frac{3}{\sqrt{13}} \approx 0.832.$$

Therefore, $\gamma_1 > \gamma_2$, so h_1 has the larger margin.

2.4 Part (d)

I used the following code to fit the linear SVM:

```
% Data from Figure 2
X = [2 6;
     3 4;
     2 2.5;
     6 2;
     7 6;
     8 4];
y = [ 1; 1; 1; -1; -1; -1];

cvx_begin quiet
    variables w(2) b
    minimize( sum_square(w) )
    subject to
        y .* (X*w + b) >= 1;
cvx_end

m_svm = -w(1)/w(2);
c_svm = -b/w(2);
fprintf('SVM boundary: %.2f x1 - x2 + %.2f = 0\n', m_svm, c_svm);
```

which gives the following output:

SVM boundary: 4.00 x1 - x2 + -15.00 = 0

This boundary is closer to the hyperplane h_1 than to h_2 , since its slope is nearer to that of h_1 .

3 Problem 3

Given

$$K_1(p, q) = (1 + p^\top q)^2, \quad K_2(p, q) = (4 + 3p^\top q)^2.$$

Expand both kernels with $t = p^\top q$:

$$K_1 = 1 + 2t + t^2, \quad K_2 = 16 + 24t + 9t^2.$$

Since K_2 is *not* a constant multiple of K_1 , there is no universal scalar mapping $\beta_i = c \alpha_i$ that always converts the optimizer of one dual problem to the other.

Therefore, in general, β must be solved from the dual optimization with kernel K_2 directly (it cannot be written only as a data-independent closed-form function of α).

4 Problem 4

Consider the dataset:

A_1	A_2	A_3	y
1	1	5	+
1	1	7	+
1	0	8	−
0	0	3	+
0	1	7	−
0	1	4	−
0	0	5	−
1	0	6	+
0	1	1	−

4.1 Part (a)

$$\Pr(y = +) = \frac{4}{9}.$$

$$\Pr(A_2 = 1 \mid y = -) = \frac{3}{5}.$$

4.2 Part (b)

Given $p = (1, 0, 7)$,

Full Bayes classifier:

In the training data, the configuration $(1, 0, 7)$ does not appear in either class; therefore

$$\Pr(p \mid y = +) = 0, \quad \Pr(p \mid y = -) = 0.$$

Hence, with a plain frequency estimate (no smoothing), both posterior scores are zero and the Full Bayes classifier is *undetermined* for this sample.

Naive Bayes classifier:

LHS:

$$\Pr(A_1 = 1 \mid y = +) = \frac{3}{4}$$

$$\Pr(A_2 = 0 \mid y = +) = \frac{2}{4}$$

$$\Pr(A_3 = 7 \mid y = +) = \frac{1}{4}$$

$$\Pr(x = \mathbf{p} \mid y = +) = \frac{3}{4} * \frac{2}{4} * \frac{1}{4} = \frac{3}{32}$$

RHS:

$$\Pr(A_1 = 1 \mid y = -) = \frac{1}{5}$$

$$\Pr(A_2 = 0 \mid y = -) = \frac{2}{5}$$

$$\Pr(A_3 = 7 \mid y = -) = \frac{1}{5}$$

$$\Pr(x = \mathbf{p} \mid y = -) = \frac{1}{5} * \frac{2}{5} * \frac{1}{5} = \frac{2}{125}$$

LHS $\hat{=}$ RHS, therefore Naive Bayes predicts $y = -$.

4.3 Part (c)

For each training point $x^{(i)} = (A_1, A_2, A_3)$,

$$d_i = \|x^{(i)} - p\|_2 = \sqrt{(A_1 - 1)^2 + (A_2 - 0)^2 + (A_3 - 7)^2}.$$

Training point $x^{(i)}$	$y^{(i)}$	d_i
(1, 1, 5)	+	$\sqrt{5} \approx 2.236$
(1, 1, 7)	+	1
(1, 0, 8)	-	1
(0, 0, 3)	+	$\sqrt{17} \approx 4.123$
(0, 1, 7)	-	$\sqrt{2} \approx 1.414$
(0, 1, 4)	-	$\sqrt{11} \approx 3.317$
(0, 0, 5)	-	$\sqrt{5} \approx 2.236$
(1, 0, 6)	+	1
(0, 1, 1)	-	$\sqrt{38} \approx 6.164$

The three closest points are those with distance 1:

$$(1, 1, 7) (+), \quad (1, 0, 8) (-), \quad (1, 0, 6) (+).$$

Their labels are +, -, +, so majority voting gives $\hat{y} = +$.

4.4 Part (d)

Assume the two classes follow individual multivariate Gaussian distributions

$$\mathcal{N}_c(\mu_c, \Sigma_c), \quad c = +, -.$$

$$\mathcal{D}_+ = \{(1, 1, 5), (1, 1, 7), (0, 0, 3), (1, 0, 6)\}, \quad \mathcal{D}_- = \{(1, 0, 8), (0, 1, 7), (0, 1, 4), (0, 0, 5), (0, 1, 1)\}.$$

Finding sample means and covariance matrices of each class:

$$\hat{\mu}_+ = (0.75, 0.50, 5.25), \quad \hat{\mu}_- = (0.20, 0.60, 5.00).$$

$$\hat{\Sigma}_+ = \begin{bmatrix} 0.1875 & 0.1250 & 0.5625 \\ 0.1250 & 0.2500 & 0.3750 \\ 0.5625 & 0.3750 & 2.1875 \end{bmatrix}, \quad \hat{\Sigma}_- = \begin{bmatrix} 0.1600 & -0.1200 & 0.6000 \\ -0.1200 & 0.2400 & -0.6000 \\ 0.6000 & -0.6000 & 6.0000 \end{bmatrix}.$$

Priors from class frequencies:

$$\Pr(y = +) = \frac{4}{9}, \quad \Pr(y = -) = \frac{5}{9}.$$

For class $c \in \{+, -\}$, work in log domain (same argmax as $\Pr(y = c) f_c(q)$ on the slide):

$$g_c(q) = \log \Pr(y = c) + \log f_c(q),$$

$$g_c(q) = \log \Pr(y = c) - \frac{1}{2} \log |\hat{\Sigma}_c| - \frac{1}{2} (q - \hat{\mu}_c)^\top \hat{\Sigma}_c^{-1} (q - \hat{\mu}_c).$$

Given $q = (0.88, -0.21, 7.89)$, we have

$$g_+(q) \approx -8.500, \quad g_-(q) \approx -3.865.$$

Since $g_-(q) > g_+(q)$, the Bayesian Gaussian classifier predicts

$$\hat{y} = -.$$

5 Problem 5

We use the **word occurrence counts** from the previous homework:

Page	Technology	Speed	Death	Physics	Olympic	Rival	Champion
Stephen Hawking	18	2	46	107	0	0	1
James Watson	8	0	7	6	0	0	0
Tu Youyou	18	0	1	4	2	0	0
Albert Einstein	12	7	19	189	0	0	0
Timothy Berners-Lee	34	0	0	3	1	0	0
Kobe Bryant	1	1	62	0	38	1	6
Cristiano Ronaldo	0	1	4	0	7	1	3
Lionel Messi	0	3	1	0	15	12	19
Michael Phelps	1	1	2	0	326	5	11
Usain Bolt	9	24	0	3	177	8	26
Total	101	39	142	312	566	27	66

Let $X \in \mathbb{R}^{10 \times 7}$ have rows $x^{(i)\top}$, where $x^{(i)} \in \mathbb{R}^7$ is the count vector for page i (same row order as the HW2 table):

$$X = \begin{bmatrix} 18 & 2 & 46 & 107 & 0 & 0 & 1 \\ 8 & 0 & 7 & 6 & 0 & 0 & 0 \\ 18 & 0 & 1 & 4 & 2 & 0 & 0 \\ 12 & 7 & 19 & 189 & 0 & 0 & 0 \\ 34 & 0 & 0 & 3 & 1 & 0 & 0 \\ 1 & 1 & 62 & 0 & 38 & 1 & 6 \\ 0 & 1 & 4 & 0 & 7 & 1 & 3 \\ 0 & 3 & 1 & 0 & 15 & 12 & 19 \\ 1 & 1 & 2 & 0 & 326 & 5 & 11 \\ 9 & 24 & 0 & 3 & 177 & 8 & 26 \end{bmatrix}.$$

Step A (labels).

$$y^{(i)} = \begin{cases} +1, & i = 1, \dots, 5 \quad (\text{scientist}), \\ -1, & i = 6, \dots, 10 \quad (\text{athlete}). \end{cases}$$

Step B (row normalization). With $\varepsilon = 10^{-6}$,

$$\tilde{x}^{(i)} = \frac{x^{(i)}}{\|x^{(i)}\|_2 + \varepsilon}.$$

Calculating

i	Page	$\ x^{(i)}\ _2$	$\tilde{x}^{(i)}$						
1	Hawking	117.872813	0.152707	0.016967	0.390251	0.907758	0	0	0.008484
2	Watson	12.206556	0.655385	0	0.573462	0.491539	0	0	0
3	Tu Youyou	18.574176	0.969087	0	0.053838	0.215353	0.107676	0	0
4	Einstein	190.459970	0.063005	0.036753	0.099758	0.992335	0	0	0
5	Berners-Lee	34.146742	0.995703	0	0	0.087856	0.029285	0	0
6	Kobe	72.986300	0.013701	0.013701	0.849474	0	0.520646	0.013701	0.082207
7	Ronaldo	8.717798	0	0.114708	0.458831	0	0.802955	0.114708	0.344124
8	Messi	27.202941	0	0.110282	0.036761	0	0.551411	0.441129	0.698454
9	Phelps	326.233046	0.003065	0.003065	0.006131	0	0.999286	0.015326	0.033718
10	Bolt	180.928163	0.049743	0.132649	0	0.016581	0.978289	0.044216	0.143703

Step C (de-meaning).

$$\mu_X = \frac{1}{10} \sum_{i=1}^{10} \tilde{x}^{(i)}, \quad g^{(i)} = \tilde{x}^{(i)} - \mu_X.$$

Numerically (to six decimals), the mean of the normalized rows is

$\mu_X^\top$						
Tech.	Speed	Death	Phys.	Olympic	Rival	Champ.
0.290240	0.042813	0.246851	0.271142	0.398955	0.062908	0.131069

and the de-meaned vectors $g^{(i)}$ are:

i	Page	$g^{(i)}$						
1	Hawking	-0.137533	-0.025845	0.143400	0.636616	-0.398955	-0.062908	-0.122585
2	Watson	0.365146	-0.042813	0.326612	0.220397	-0.398955	-0.062908	-0.131069
3	Tu Youyou	0.678848	-0.042813	-0.193013	-0.055789	-0.291278	-0.062908	-0.131069
4	Einstein	-0.227234	-0.006060	-0.147092	0.721192	-0.398955	-0.062908	-0.131069
5	Berners-Lee	0.705463	-0.042813	-0.246851	-0.183286	-0.369669	-0.062908	-0.131069
6	Kobe	-0.276539	-0.029111	0.602624	-0.271142	0.121691	-0.049207	-0.048862
7	Ronaldo	-0.290240	0.071895	0.211981	-0.271142	0.404000	0.051800	0.213055
8	Messi	-0.290240	0.067470	-0.210090	-0.271142	0.152456	0.378221	0.567385
9	Phelps	-0.287174	-0.039747	-0.240720	-0.271142	0.600331	-0.047582	-0.097351
10	Bolt	-0.240496	0.089837	-0.246851	-0.254561	0.579334	-0.018692	0.012634

We stack the $g^{(i)\top}$ as rows to form $G \in \mathbb{R}^{10 \times 7}$.

5.1 Parts (a)–(c): MATLAB outputs

Numerical results from `problem5.m` for the perceptron, linear SVM, and Gaussian Bayes classifiers.

Part (a) Perceptron

Standard perceptron with at most 1000 iterations yields

$$w_p^\top = [0.541 \quad -0.069 \quad -0.050 \quad 0.581 \quad -0.690 \quad -0.126 \quad -0.254], \quad \gamma_p = 0.366.$$

Part (b) Linear SVM (CVX)

Hard-margin linear SVM gives

$$w_s^\top = [1.116 \quad -0.049 \quad -0.406 \quad 1.179 \quad -1.078 \quad -0.152 \quad -0.344], \quad b_s = -0.022, \quad \gamma_s = 0.494.$$

Part (c) Gaussian Bayes (equal priors, ridge on Σ_c)

Ridge parameter $\lambda = 10^{-3}$; we set $\Sigma_c \leftarrow \Sigma_c + \lambda I$ after computing the sample covariances. Estimated class means (column vectors in $\mathbb{R}^7$):

$$\mu_+ = \begin{bmatrix} 0.277 \\ -0.032 \\ -0.023 \\ 0.268 \\ -0.372 \\ -0.063 \\ -0.129 \end{bmatrix}, \quad \mu_- = \begin{bmatrix} -0.277 \\ 0.032 \\ 0.023 \\ -0.268 \\ 0.372 \\ 0.063 \\ 0.129 \end{bmatrix}.$$

Regularized covariance matrices:

$$\Sigma_+ = \begin{bmatrix} 0.157 & -0.005 & -0.028 & -0.142 & 0.011 & 0.000 & -0.001 \\ -0.005 & 0.001 & 0.000 & 0.005 & 0.000 & 0.000 & 0.000 \\ -0.028 & 0.000 & 0.050 & 0.029 & -0.005 & 0.000 & 0.000 \\ -0.142 & 0.005 & 0.029 & 0.131 & -0.010 & 0.000 & 0.001 \\ 0.011 & 0.000 & -0.005 & -0.010 & 0.003 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.001 & 0.000 \\ -0.001 & 0.000 & 0.000 & 0.001 & 0.000 & 0.000 & 0.001 \end{bmatrix},$$

$$\Sigma_- = \begin{bmatrix} 0.001 & 0.000 & -0.001 & 0.000 & 0.002 & -0.001 & -0.002 \\ 0.000 & 0.004 & -0.007 & 0.000 & 0.001 & 0.004 & 0.008 \\ -0.001 & -0.007 & 0.115 & -0.001 & -0.041 & -0.018 & -0.020 \\ 0.000 & 0.000 & -0.001 & 0.001 & 0.001 & 0.000 & 0.000 \\ 0.002 & 0.001 & -0.041 & 0.001 & 0.042 & -0.017 & -0.025 \\ -0.001 & 0.004 & -0.018 & 0.000 & -0.017 & 0.027 & 0.038 \\ -0.002 & 0.008 & -0.020 & 0.000 & -0.025 & 0.038 & 0.060 \end{bmatrix}.$$

Training error on (G, y) is 0.000 (0/10 misclassified).

5.2 Part (d) Testing

By counting occurrences of the keywords on each page, we obtain the following raw count vectors:

Page	Tech.	Speed	Death	Phys.	Olympic	Rival	Champ.
Niels Bohr ($x^{(11)}$)	3	2	3	64	2	0	0
Ariel Ortega ($x^{(12)}$)	0	0	0	0	8	0	0
Winston Churchill ($x^{(13)}$)	1	0	17	1	1	1	1

Test preprocessing (same as training). With the **same** $\varepsilon = 10^{-6}$ and **same** training mean μ_X from Step C above,

$$\tilde{x}^{(j)} = \frac{x^{(j)}}{\|x^{(j)}\|_2 + \varepsilon}, \quad g^{(j)} = \tilde{x}^{(j)} - \mu_X, \quad j \in \{11, 12, 13\}.$$

Numerically,

j	Page	$\ x^{(j)}\ _2$	$\tilde{x}^{(j)}$						
11	Bohr	64.202804	0.046727	0.031151	0.046727	0.996841	0.031151	0	0
12	Ortega	8	0	0	0	0	1	0	0
13	Churchill	17.146428	0.058321	0	0.991460	0.058321	0.058321	0.058321	0.058321

j	Page	$g^{(j)}$							
11	Bohr	-0.243513	-0.011662	-0.200124	0.725699	-0.367804	-0.062908	-0.131069	
12	Ortega	-0.290240	-0.042813	-0.246851	-0.271142	0.601045	-0.062908	-0.131069	
13	Churchill	-0.231919	-0.042813	0.744609	-0.212821	-0.340634	-0.004587	-0.072748	

Predicted labels. We apply each $g^{(j)}$ to the three classifiers from parts (a)–(c). Using `problem5.m` (CVX for the linear SVM, with ridge $\lambda = 10^{-3}$ for Bayes), we obtain:

Test page	Perceptron $\hat{y}$	Linear SVM $\hat{y}$	Gaussian Bayes $\hat{y}$
Niels Bohr	+1	+1	+1
Ariel Ortega	-1	-1	-1
Winston Churchill	-1	-1	-1

(Encode +1 as scientist and -1 as athlete.)

Interpretation. Bohr is classified as a scientist because his counts emphasize *physics*. Ortega is classified as an athlete because only the *Olympic* count is nonzero, similar to the athlete training pages. Churchill is labeled as an athlete even though he was not one: he was a politician but his counts (e.g. *death*) resemble the athlete side of the training data more than the scientist side. Maybe the keywords choice is not good for this task.

6 Appendix: Problem 5 (MATLAB code and output)

The CVX package must be on the MATLAB path for the linear SVM. The helper files `perceptron.m` and `geometric_margin.m` are in the same folder as `problem5.m`.

Source code

```
problem5.m

clear; clc;

X = [
    18  2 46 107   0  0  1
     8  0  7   6   0  0  0
    18  0  1   4   2  0  0
    12  7 19 189   0  0  0
    34  0  0   3   1  0  0
     1  1 62   0  38  1  6
     0  1  4   0   7  1  3
     0  3  1   0  15 12 19
     1  1  2   0 326  5 11
     9 24  0   3 177  8 26
];

y = [ones(5, 1); -ones(5, 1)];

eps_norm = 1e-6;
row_norm = sqrt(sum(X.^2, 2)) + eps_norm;
X_tilde = X ./ row_norm;
mu_X = mean(X_tilde, 1);
G = X_tilde - mu_X;

[N, d] = size(G);

% --- (a) Perceptron ---
T = 1000;
w_p_row = perceptron(G, y, T);
w_p = w_p_row(:);

gamma_p = geometric_margin(G, y, w_p, 0);

fprintf('(a) Perceptron\n');
fprintf('    w_p^T = [');
fprintf('%.3f ', w_p');
fprintf(']\n');
fprintf('    gamma_p = %.3f\n\n', gamma_p);

% --- (b) Linear SVM ---

cvx_begin quiet
    variables w_s(d) b_s
    minimize( 0.5 * (w_s' * w_s) )
    subject to
        y .* (G * w_s + b_s) >= 1;
cvx_end
```

```

gamma_s = geometric_margin(G, y, w_s, b_s);

fprintf('(b) Linear SVM \n');
fprintf('    w_s^T = [');
fprintf('%.3f ', w_s');
fprintf(']\n');
fprintf('    b_s = %.3f\n', b_s);
fprintf('    gamma_s = %.3f\n\n', gamma_s);

% --- (c) Gaussian Bayes, equal priors 1/2, ridge on covariances ---
lambda_ridge = 1e-3;
fprintf('(c) Gaussian Bayes (ridge lambda = %.3f)\n', lambda_ridge);

idx_pos = y == 1;
idx_neg = y == -1;
G_pos = G(idx_pos, :);
G_neg = G(idx_neg, :);

mu_plus = mean(G_pos, 1)';
mu_minus = mean(G_neg, 1)';

%regularization
Sigma_plus = cov(G_pos, 1) + lambda_ridge * eye(d);
Sigma_minus = cov(G_neg, 1) + lambda_ridge * eye(d);

fprintf('    mu_+ =\n    ');
fprintf('%.3f ', mu_plus');
fprintf('\n');
fprintf('    Sigma_+ =\n');
print_matrix_3dp(Sigma_plus);
fprintf('    mu_- =\n    ');
fprintf('%.3f ', mu_minus');
fprintf('\n');
fprintf('    Sigma_- =\n');
print_matrix_3dp(Sigma_minus);

log_prior = log(0.5);
y_hat = zeros(N, 1);
for i = 1:N
    g = G(i, :)';
    ll_plus = log_prior + logmvnpdf_col(g, mu_plus, Sigma_plus);
    ll_minus = log_prior + logmvnpdf_col(g, mu_minus, Sigma_minus);
    if ll_plus >= ll_minus
        y_hat(i) = 1;
    else
        y_hat(i) = -1;
    end
end

train_err = mean(y_hat ~= y);
fprintf('    Training error = %.3f (%d / %d misclassified)\n', train_err, sum(y_hat ~= y), N);

% --- Local helper: print matrix with 3 decimal places ---

```

```

function print_matrix_3dp(M)
    [nr, nc] = size(M);
    for r = 1:nr
        fprintf(' ');
        fprintf('%3f ', M(r, :));
        fprintf('\n');
    end
end

% --- Local helper: log-density of N(mu, Sigma), x and mu column vectors ---
function lp = logmvnpdf_col(x, mu, Sigma)
    dloc = length(x);
    xc = x - mu;
    R = chol(Sigma, 'lower');
    log_det = 2 * sum(log(diag(R)));
    z = R \ xc;
    quad = z' * z;
    lp = -0.5 * (dloc * log(2 * pi) + log_det + quad);
end

```

perceptron.m

```

function w = perceptron(X, y, T)

    [N, d] = size(X);
    w = zeros(1, d);
    iter = 1;

    while iter <= T
        found = false;
        for i = 1:N
            if y(i) * (w * X(i, :))' <= 0
                w = w + y(i) * X(i, :);
                found = true;
                break;
            end
        end

        if ~found
            break;
        end
        iter = iter + 1;
    end
end

```

geometric_margin.m

```

function gamma = geometric_margin(G, y, w, b)

    if nargin < 4 || isempty(b)
        b = 0;
    end
    w = w(:);

```

```

    denom = norm(w);
    if denom == 0
        gamma = NaN;
        return
    end
    s = y(:) .* (G * w + b);
    gamma = min(s) / denom;
end

```

Program output

Console output from running `problem5.m` (three decimal places):

(a) Perceptron

```

w_p^T = [0.541 -0.069 -0.050 0.581 -0.690 -0.126 -0.254 ]
gamma_p = 0.366

```

(b) Linear SVM

```

w_s^T = [1.116 -0.049 -0.406 1.179 -1.078 -0.152 -0.344 ]
b_s = -0.022
gamma_s = 0.494

```

(c) Gaussian Bayes (ridge lambda = 0.001)

```

mu_+ =
0.277 -0.032 -0.023 0.268 -0.372 -0.063 -0.129
Sigma_+ =
0.157 -0.005 -0.028 -0.142 0.011 0.000 -0.001
-0.005 0.001 -0.000 0.005 -0.000 0.000 0.000
-0.028 -0.000 0.050 0.029 -0.005 0.000 0.000
-0.142 0.005 0.029 0.131 -0.010 0.000 0.001
0.011 -0.000 -0.005 -0.010 0.003 0.000 -0.000
0.000 0.000 0.000 0.000 0.000 0.001 0.000
-0.001 0.000 0.000 0.001 -0.000 0.000 0.001
mu_- =
-0.277 0.032 0.023 -0.268 0.372 0.063 0.129
Sigma_- =
0.001 0.000 -0.001 0.000 0.002 -0.001 -0.002
0.000 0.004 -0.007 0.000 0.001 0.004 0.008
-0.001 -0.007 0.115 -0.001 -0.041 -0.018 -0.020
0.000 0.000 -0.001 0.001 0.001 -0.000 -0.000
0.002 0.001 -0.041 0.001 0.042 -0.017 -0.025
-0.001 0.004 -0.018 -0.000 -0.017 0.027 0.038
-0.002 0.008 -0.020 -0.000 -0.025 0.038 0.060
Training error = 0.000 (0 / 10 misclassified)

```